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The Harrow-Hassidim-Lloyd algorithm

PhD course "The ABCs of Quantum Computing"

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Let $Ax = b$ be a linear system with A $N_b \times N_b$ matrix and x and b N_b -dimensional vectors.

Matrix A can be written a linear combination of outer products of its eigenvectors weighted by its eigenvalues:

$$A = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$$

The inverse matrix A^{-1} can be expressed as

$$A^{-1} = \sum_i \lambda_i^{-1} \mathbf{u}_i \mathbf{u}_i^\top$$

1. State preparation
2. Quantum Phase Estimation (QPE)
3. Ancilla qubit rotation
4. Inverse QPE

Little endian convention (rightmost qubit is the less significant). Then, from the most significant to the least significant

1. *b-register*: components of $\mathbf{x}, \mathbf{b}$, encoded as amplitudes/coefficients of basis states of the n_b -qubits, with $N_b = 2^{n_b}$
2. *c-register*: values of the phase of the eigenvalues of A after QPE
3. *ancilla*: used for inversion and measurement, discarded in the solution

Initial state:

$$|\psi_0\rangle = |0\rangle^{\otimes n_b} |0\rangle^{\otimes n} |0\rangle$$

ancilla qubit _____
c-registers _____
b-registers _____

The state preparation step consists of rotating the qubits in the b -register to have amplitudes corresponding to the coefficients of $\mathbf{b}$ (*amplitude encoding*)

$$|b\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle + \cdots + \beta_{N_b-1} |N_b - 1\rangle$$

Leading to the updated state:

$$|\psi_1\rangle = |b\rangle_b |0\rangle_a^{\otimes n}$$

The amplitude encoding is implemented in the following way:

1. $\mathbf{b}$ is normalized: $\tilde{\mathbf{b}}$ (amplitudes need to sum to 1 in quantum states)
2. $|\tilde{b}\rangle = \tilde{b}_0 |0\rangle + \tilde{b}_1 |1\rangle_1 + \cdots + \tilde{b}_{N_b-1} |N_b - 1\rangle$

Consider the simple example where $\mathbf{b} = (0, 1)$. The vector is already normalized and the state preparation circuit corresponds to the following:

Note: The state preparation step is then specific to each $\mathbf{b}$.

The goal of QPE is to estimate the phase of the eigenvalues of A

Overview:

1. Superposition
2. Controlled rotations
3. Inverse Quantum Fourier Transforms (IQFT)

The goal of this step is to put the clock qubits into an equal superposition and explore all possible phases simultaneously

$$\begin{aligned} |\Psi_2\rangle &= I^{\otimes n_b} \otimes H^{\otimes n} \otimes I |\Psi_1\rangle \\ &= |b\rangle \frac{1}{2^{\frac{n}{2}}} (|0\rangle + |1\rangle)^{\otimes n} |0\rangle \end{aligned}$$

In order to estimate the eigenvalues of A , we need it to somehow operate with it within the quantum circuit. As A is non-unitary and gates of a quantum circuit need to be unitary we have to encode A through Hamiltonian encoding, namely we're gonna use the unitary gate

$$U = e^{iAt}$$

Interpretations:

- eigenvalue connection: $\text{eig}(U) = e^{i\lambda_i t}$, where λ_i are the eigenvalues of A
- A energy of the Hamiltonian system

Eigenvalues of A will be stored in the c -registers, encoded using *basis encoding*. For this reason, the stored eigenvalues are not exact. In this section, controlled U^t gates will be applied to $|b\rangle$.

Motivation: Assume $|b\rangle$ to be an eigenvector of U with eigenvalues $e^{2\pi i\phi}$. Then

$$U|b\rangle = e^{2\pi i\phi}|b\rangle.$$

When the control qubit is 0 $|b\rangle$ is not affected, otherwise U is applied to it. This relationship is expressed by

$$\begin{aligned} |\Psi_3\rangle &= |b\rangle \otimes \left(\frac{1}{2^{\frac{n}{2}}} \bigotimes_{j=1}^n (|0\rangle + e^{2\pi i\psi 2^{n-j}} |1\rangle) \right) \otimes |0\rangle_a \\ &= |b\rangle \otimes \left(\frac{1}{2^{\frac{n}{2}}} \sum_{k=1}^{2^n-1} e^{2\pi i\psi k} |k\rangle \right) \otimes |0\rangle_a \end{aligned}$$

$$\begin{aligned}
|\Psi_4\rangle &= |b\rangle \otimes \text{IQFT}\left(\frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} (e^{2\pi i \psi k} |k\rangle)\right) \otimes |0\rangle_a \\
&= |b\rangle \otimes \left(\frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} (e^{2\pi i \psi k} \text{IQFT}(|k\rangle))\right) \otimes |0\rangle_a \\
&= |b\rangle \otimes \left(\frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} (e^{2\pi i \psi k} \left(\frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-2\pi i j k / N} |j\rangle\right))\right) \otimes |0\rangle_a \\
&= |b\rangle \otimes \left(\frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} (e^{2\pi i \psi k} \left(\frac{1}{\sqrt{2^n}} \sum_{j=0}^{N-1} e^{-2\pi i j k / N} |j\rangle\right))\right) \otimes |0\rangle_a \\
&= |b\rangle \otimes \left(\frac{1}{2^n} \sum_{k=0}^{2^n-1} \left(\sum_{j=0}^{2^n-1} e^{2\pi i k (\phi - j/N)} |j\rangle\right)\right) \otimes |0\rangle_a
\end{aligned}$$

Thus,

$$|\Psi_4\rangle = |b\rangle \otimes \left(\frac{1}{2^n} \sum_{k=0}^{2^n-1} \left(\sum_{j=0}^{2^n-1} e^{2\pi i k(\phi-j/N)} |j\rangle \right) \right) \otimes |0\rangle_a$$

and the following alternatives apply:

- if $\phi - j/N = 0$: $|j\rangle$ will have finite amplitude equal to $\sum_{k=0}^{2^n-1} e^0 = 2^n$, *constructive interference*
- otherwise: $\sum_{k=0}^{2^n-1} e^{2\pi i k(\phi-j/N)} = 0$, *destructive interference*

Finally, ignoring the states with zero-amplitude:

$$\begin{aligned} |\Psi_4\rangle &= \frac{1}{2^n} |b\rangle \sum_{k=0}^{2^n-1} e^{2\pi i k \cdot 0} |N\phi\rangle |0\rangle_a \\ &= |b\rangle |N\phi\rangle |0\rangle_a \end{aligned}$$

In this way, the clock qubits are used to represent the phase information of U , the accuracy depends on the number of qubits n

Since U encodes A through Hamiltonian encoding $U = e^{iAt}$, U is diagonal in the eigenvectors of A :

$$U|b\rangle = e^{i\lambda_j t} |u_j\rangle$$

Equating the latter to $U|b\rangle = e^{2\pi i\phi} |b\rangle$, we get $\phi = \frac{\lambda_j t}{2\pi}$ and

$$|\Psi_4\rangle = |u_j\rangle \left| \frac{N\lambda_j t}{2\pi} \right\rangle |0\rangle_a$$

More generally, expressing $|b\rangle$ in the basis formed by the eigenvectors of A , we have

$$\sum_{j=0}^{2^n-1} b_j |u_j\rangle \left| \frac{N\lambda_j t}{2\pi} \right\rangle |0\rangle_a$$

As λ_j are usually non-integers, t will be selected so that $\frac{N\lambda_j t}{2\pi}$ are integers

Rotation of the ancilla qubit is performed on the basis of a parameter C

$$|\psi_5\rangle = \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle |\tilde{\lambda}_j\rangle \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle_a + \frac{C}{\tilde{\lambda}_j} |1\rangle_a \right)$$

Once the ancilla qubit is measured we got two alternatives

- if $|0\rangle$: repeat the computation
- if $|1\rangle$: stop the iterations

After such iterations, the state is updated to

$$|\psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle |\tilde{\lambda}_j\rangle \frac{C}{\tilde{\lambda}_j} |1\rangle_a$$

$$|\Psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle |\tilde{\lambda}_j\rangle |1\rangle_a$$

The *b-register* now stores the result in the desired form with respect to the eigenvector-basis. What we have to do now is to express the *b-register* through basis encoding. Problem is that the latter is entangled with the other qubits making unfeasible to operate such a conversion. The inverse QPE step allows then to disentangle the states

allowing to express the *b-register* through basis encoding.

To do so, QFT is applied to the *c-register* qubits:

$$\begin{aligned}
 |\Psi_7\rangle &= \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \text{QFT}(|\tilde{\lambda}_j\rangle) |1\rangle_a \\
 &= \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \left(\frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{2\pi i y \tilde{\lambda}_j / N} |y\rangle \right) |1\rangle_a
 \end{aligned}$$

Then controlled rotations with U^{-1} are applied as before

$$\begin{aligned}
 |\psi_8\rangle &= \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \left(\frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{-i\lambda_j t y} e^{2\pi i y \tilde{\lambda}_j / N} |y\rangle \right) |1\rangle_a \\
 &= \frac{1}{2^{\frac{n}{2}} \sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \sum_{y=0}^{2^n-1} |y\rangle |1\rangle_a \\
 &= \frac{C}{2^{\frac{n}{2}} \sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} |x\rangle \sum_{y=0}^{2^n-1} \frac{1}{2^{\frac{n}{2}}} |y\rangle |1\rangle_a
 \end{aligned}$$

The clock-qubits and *b-registers* are now unentangled. By applying the Hadamard gates on the clock qubits we have

$$|\psi_9\rangle = \frac{C}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N}{\tilde{\lambda}_j} \right|^2} |x\rangle_b |0\rangle_c^{\otimes n} |1\rangle_a$$

where the solution $|x\rangle$ is stored in the *b-registers* successfully.